

Converting from Customary to Metric

$$1 \text{ ounce} = 28.4 \text{ grams}$$

Within the Customary System

$$1 \text{ pound} = 16 \text{ ounces}$$

Converting from Metric to Customary

$$1 \text{ kilogram} = 2.2 \text{ pounds}$$

Within the Metric System

$$1 \text{ kilogram} = 1000 \text{ grams}$$

A recipe was written in metric units. Determine the amount of each ingredient in U.S. Customary measurements. Round to the nearest thousandth.

1. 260 grams light brown sugar = 9.155 ounces

$$260 \div 28.4 = 9.155$$

2. 0.575 kilograms of flour = 1.265 pounds

$$2.2 \cdot 0.575 = 1.265$$

A recipe was written in US Customary measurements. Convert to metrics, rounding to the nearest thousandth.

3. 16 ounces of chocolate chips = 454.4 grams

$$16 \cdot 28.4 = 454.4$$

4. 0.75 pounds of sugar = 340.909 grams

$$\left(\frac{0.75}{2.2}\right) \cdot 1,000 = 340.909$$

5. If 1 kilogram = 1000 grams, and 1 kilogram = 2.2 pounds, how many grams is equal to 26 pounds?

$$\frac{2.2 \text{ lbs}}{1 \text{ kg}} = \frac{26 \text{ lbs}}{11.818 \text{ kg}} \quad \frac{11.818 \text{ kg}}{11,818.182 \text{ g}} = \frac{1 \text{ kg}}{1,000 \text{ g}} = 11,818.182 \text{ grams}$$

Use the conversions below for 6-8. Round to the nearest thousandth.

$$1 \text{ ft} = 0.305 \text{ meters (m)}$$

$$10.764 \text{ square feet} = 1 \text{ square meter}$$

6. A floorplan of a house shows the office as having dimensions 12 feet $\times$ 10 feet. Find the area of the office in square meters.

$$\text{Area} = 12 \cdot 10 = 120 \text{ ft}^2$$

$$\frac{120 \text{ ft}^2}{11.148 \text{ m}^2} = \frac{10.764 \text{ ft}^2}{1 \text{ m}^2}$$

$$\text{Area} = 11.148 \text{ m}^2$$

7. The same floorplan has a closet that is 5 feet by 7 feet. What is the area in square meters?

$$5 \cdot 7 = 35 \text{ ft}^2$$

$$35 \div 10.764 = 3.252 \text{ m}^2$$

8. The master bedroom of the house is 15.6 square meters. Find the area in square feet.

$$15.6 \cdot 10.764 = 167.918 \text{ ft}^2$$

Using the unit conversions below, determine the missing values for 9-10. Round to the nearest thousandth.

$$1000 \text{ acres} = 1.5625 \text{ square miles}$$

$$100 \text{ acres} = 0.405 \text{ square kilometers}$$

9. 57 acres = 0.089 square miles

$$\frac{57 \text{ acres}}{0.089 \text{ mi}^2} = \frac{1000 \text{ acres}}{1.5625 \text{ mi}}$$

10. 92 square kilometers = 22,716.049 acres

$$\frac{92 \text{ km}^2}{22,716.049 \text{ acres}} = \frac{0.405 \text{ km}^2}{100 \text{ acres}}$$